

demonstrated chips that can control the flight of pigeons, DARPA in the US is experimenting with cyborg insects. It may be cheaper to stick a camera or a microphone on a real insect, and a chip to control its mind, rather than building tiny drones that can infiltrate enemy territories.

09 DRONE SWARMS

Swarms of drones coordinating with each other can have many military applications. These drones can provide surveillance, confuse the targeting mechanisms of enemy weapons, or help search and rescue operations. In the US, there have already been tests of what could happen if two drone swarms were made to engage with each other.

10 EXOSUITS

There have already been tests of using powered exoskeletons for soldiers. These exosuits could significantly improve the amount of equipment and supplies that a single soldier can carry, boosting their efficiency in the battlefield. Another innovative application is the Marine Mojo exoskeleton, that dampens the movement experienced by personnel on fast moving marine craft.

11 FORCE FIELDS

Many militaries around the world are working towards realising force fields. Military scientists in the UK are developing rapid response systems that can detect and fire at incoming projectiles, providing a protective bubble around a vehicle. Boeing has patented a system that uses plasma to mitigate the effects of explosions, by intercepting the resulting shockwaves in walls of temperature and air density changes.

12 LAWS

Lethal autonomous weapons systems or LAWS can include anything from weaponized drones to autonomous tanks. There are serious ethical considerations for deploying these in the battlefield, including the ability to show mercy when killing is no longer required to achieve the objectives.

13 NLOS CANNON

The non line of sight (NLOS) canon is a fire and forget type of weapon that use active guidance to takedown enemy targets, even if they are not in the line of fire. The type of acceptable targets are fed into the munitions themselves, and once they are fired they can actively maneuver themselves to the target. Most of the operations by the crew are handled autonomously, so the NLOS cannon is always ready to fire.

14 DIRECTED ENERGY WEAPONS

Think of them as railguns, or powerful lasers that can take down all kinds of targets ranging from quad-

16 SONIC WEAPONS

Devices that produce high power sound waves are already being used for riot control as they disorient people and induce nausea. In the seas, long range acoustic devices are used to deter pirates from attacking ships. In combat situations, sufficiently high power sound waves can rupture eardrums, which can incapacitate combatants. Militaries are researching sonic and ultrasonic devices, like mines, bullets and grenades.

17 XR TRAINING

VR and AR training is already being used to prepare soldiers for handling new



The Lockheed Martin Modular Active Protection System

copters to LAWS to military aircraft. Lasers have exceptional range, are not deflected by gravity, and can be fired discretely if the waves are not in the visible spectrum. Microwave weapons are in development too. Weapons that fire a stream of high energy particles are also possible. Russia, China, UK and India are developing DEWs.

15 ROBOTS

It has proven difficult to build a humanoid robot that can use the vehicles and equipment designed for humans. Robots tend to fall over when performing simple tasks. Still, efforts are ongoing to develop robots which replace humans for hazardous tasks. There is also a requirement for robotic mules that can ferry supplies and ammunition.

18 WEARABLES

Military wearables are personal computing devices designed to be modular, flexible, secure and long lasting. The devices are geared towards situational awareness and health monitoring. To make sure all the electronics on future soldiers are powered, a company, Bionic Power has created a wearable system that harnesses electricity from soldiers' movement, to recharge batteries. 